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Multi-layered security system: Integrating quantum key distribution with classical cryptography to enhance steganographic security

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ABSTRACT

This research introduces a novel cryptographic system that enhances the security of steganographic images by integrating Quantum Key Distribution (QKD) with traditional encryption. Using the E91 QKD protocol, it generates a shared secret key between parties, leveraging quantum entanglement to provide superior protection against eavesdropping. The quantum key is then hashed with the Secure Hash Algorithm (SHA) to produce a fixed-length, high-entropy key for symmetric encryption. The Advanced Encryption Standard (AES) is employed to encrypt steganographic images, which conceal critical data within digital images, adding an extra layer of security through obscurity. Experiments with images of varying resolutions (64 × 64 to 512 × 512 pixels) show consistent encryption performance. The encrypted images exhibit high randomness, with an average entropy of 7.9929. Differential attack resilience is demonstrated by a Number of Pixels Change Rate (NPCR) averaging 99.6928% and a Unified Average Changing Intensity (UACI) of 56.1549%. The average E91 protocol key generation time is 5.78 s with a key rate of 7.43 bps. Encryption and decryption times for test images are 0.00149 and 0.00175 s, respectively. This research combines quantum and classical cryptography with steganography, providing a robust security framework highly resistant to both quantum and classical threats.

1. Introduction

The rapid advancements in digital communication and data storage have necessitated the development of robust security mechanisms to protect sensitive information from unauthorized access and cyber threats. Traditional cryptographic methods have been the cornerstone of securing digital data; however, the emergence of quantum computing poses significant challenges to these classical techniques. Quantum algorithms like Shor's Algorithm [1] (It can successfully calculate factors of prime numbers with exponential speedup) and Grover's Algorithm [2] (Quantum Search Algorithm) have the potential to breach the security of classical cryptographic schemes like the RSA Algorithm [3], Ellicptic-Crurve-Cryptography(ECC) [4] and many other. Shor's algorithm exemplifies quantum advantage in prime factorization, reducing complexity to $O((\log N)^2(\log\log N)(\log\log\log N))$ through efficient quantum Fourier transforms that is far surpassing the sub-exponential $\exp(O((\log N)^{1/3}(\log\log N)^{2/3}))$ complexity of classical methods. Similarly, Grover's algorithm achieves quadratic speedup in unstructured search, completing tasks in $O(\sqrt{N})$ iterations, compared to the $O(N)$ steps required classically. These results, leveraging superposition and entanglement, highlight quantum computing's superiority in problems where classical methods are inherently limited by higher complexity bounds. Classical algorithms cannot achieve these kinds of complexity improvements. To address this problem QKD [5] presents a promising solution by utilizing the principles of quantum mechanics to enable secure key exchange, offering unparalleled security guarantees against eavesdropping. The E91 QKD [6] protocol, introduced by Ekert in 1991, is based on the phenomenon of quantum entanglement and provides a means for two parties to generate a shared secret key with absolute security assured by the laws of physics. Where two parties creates a Bell State [7], such as $\psi_{\pm} = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$. Despite its theoretical advantages, the practical implementation of QKD systems requires seamless integration with existing cryptographic infrastructures to ensure comprehensive data protection. In parallel, steganography [8] has emerged as an effective technique to conceal information within digital media, such as images, audio, and video files. By embedding data in a way that is imperceptible to the human eye, steganography adds an additional layer of security through obscurity. However, the security of steganographic methods heavily relies on the robustness of the underlying cryptographic algorithms used for data encryption [9].

Classical cryptographic systems are vulnerable to quantum algorithm [10], RSA and ECC rely on the computational difficulty of

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integer factorization and discrete logarithm problems. Shor's algorithm can solve these problems efficiently on quantum computers, rendering these cryptographic methods vulnerable. Classical key distribution relies on computational complexity assumptions. These assumptions may be invalidated by future mathematical breakthroughs or quantum computing advances. QKD offers significant security advantages over classical key distribution methods. However, classical encryption systems still provide more robust encryption than current quantum technologies [11,12]. Therefore, combining the strengths of both quantum and classical cryptography, a hybrid approach is a more practical and secure solution.

1.1. Current challenges

Despite QKD's theoretical security advantages, several practical challenges remain:

1. Since existing cryptographic infrastructure gives robust security over the current technological landscape, we need to integrate QKD with existing cryptographic infrastructure.
2. While quantum cryptography enhances security, it is constrained by technological factors such as key generation rates, noise sensitivity, and distance limitations. Developing systems that effectively combine both quantum and classical security mechanisms while maintaining optimal performance and security levels remains challenging.
3. Hybrid systems must defend against both quantum computing threats and traditional cryptographic attacks. This dual requirement complicates system design and implementation.
4. The transmission of complex data types, particularly high-resolution images, presents unique challenges in quantum–classical systems.

1.2. Proposed solution

To address these challenges, this paper presents a novel approach to hybrid quantum–classical cryptographic systems. In this paper, we proposed a framework that integrates QKD, classical encryption, and steganography for secure and efficient image transmission. The proposed solution can be summarized as follows:

1. **Secure key generation:** The E91 protocol is utilized to generate a quantum-secure shared key. This key is hashed using the Secure Hash Algorithm (SHA) to produce a high-entropy, fixed-length key suitable for symmetric encryption.
2. **Strong data encryption:** The Advanced Encryption Standard (AES) is employed to encrypt steganographic images, ensuring robust protection for the hidden data while maintaining compatibility with classical cryptographic infrastructure.
3. **Hybrid security integration:** By combining QKD with classical encryption, the system ensures resilience against both quantum-era and traditional attacks, providing a comprehensive defense against diverse threat vectors.
4. **Steganography for concealment:** Steganography adds a layer of invisibility, embedding encrypted data within cover images to obscure its presence and further enhance security against detection.

Our hybrid approach integrates quantum and classical cryptography with steganography to overcome individual limitations, enabling scalable, secure transmission of sensitive data. It advances cryptographic research by demonstrating the practical benefits of hybrid protocols and lays the foundation for future secure communication technologies, offering robust protection against evolving cyber threats.

In this paper, we have provided a detailed description of our system architecture, including the key generation process using the E91 QKD protocol, the hashing mechanism, and the encryption process. We have also conducted a comprehensive security analysis to evaluate the resilience of our system against various attack vectors and performed extensive performance evaluations to assess the efficiency of our approach. Our results demonstrate that the proposed system offers a robust and efficient solution for secure digital communication, making it suitable for applications that require stringent security measures.

1.3. Paper organization

The remainder of this paper is organized as follows: Section 2 reviews existing developments in quantum–classical cryptography and steganography. Section 3 details our proposed architecture and implementation methodology. Section 4 presents experimental results and security analysis. Section 5 discusses our findings, and Section 6 concludes with future research directions.

2. Existing developments

QKD combined with classical cryptography presents a promising frontier to enhance steganographic security. Here, we will explore the state-of-the-art research and methodologies that have integrated QKD with classical cryptographic techniques and their application in steganography, providing an extensive overview of recent advancements and their implications.

Classical cryptography refers to encryption techniques that rely on mathematical algorithms and computational complexity to secure data. It is based on established mathematical principles and has been the foundation of secure communication for decades. Classical cryptographic methods are divided into different algorithms such as Symmetric key algorithms- Advanced Encryption System(AES) [13], Data Encryption Standard (DES) [14], Asymmetric key algorithms- Rivest–Shamir–Adleman(RSA) [3], Elliptic Curve Cryptography (ECC) [4] and Hash Functions [15]. Among these, ECC enhances the security of open communication networks. A recent survey investigates scientific concepts, innovative methodologies, and implementations of ECC. ECC-based schemes are considered to be more secure for cloud computing, e-health, and e-voting than RSA and Diffie–Hellman schemes. It is also proven to be more advanced in distributed and asynchronous networking, providing significant security advantages over RSA [16]. [17] presents an ECC-based novel image encryption methodology for grayscale and color images. It employs a combination of primitive polynomials, invertible Suslin matrices, and logistic chaotic maps. This scheme aims to enhance security and efficiency in image encryption, making it robust against various attacks and suitable for secure image transmission. [18] creates a hybrid cryptographic framework to secure data storage in cloud computing, where they integrate AES and OTP with the RSA algorithm to distribute keys.

Another chaotic-based image encryption scheme from [19] combines Arnold's cat map for pixel shuffling, ECC for pixel value encryption using public and private keys, and a genetic algorithm to optimize key generation. This integration enhances data security by introducing chaos, robust encryption, and optimized key management. The approach indicates strong resistance against statistical attacks. Additionally, the extensive key space enhances resilience against brute force attacks. Research conducted by [20] combines QKD with logistic chaotic maps to establish a secure image transmission scheme. It utilizes the E91 QKD protocol for key distribution and logistic chaotic maps for encryption and decryption. The scheme also incorporates an eavesdropping detection mechanism based on CHSH inequality.

QKD uses the principles of quantum mechanics to exchange the key, which is theoretically assumed to be secure even for quantum hackers and other more sophisticated attackers. Some research works have discussed the enhancement of QKD with other classical cryptography algorithms for practicality in defense An approach of incorporating

QKD into hybrid quantum–classical networks highlighted some methods to integrate QKD into existing Transport Layer security (TLS) protocols, enhancing security for data transmission over optical fibers. It used the Post Quantum Cryptography (PQC) algorithms to secure the delivery of QKD keys, demonstrating the complementing strengths of QKD and PQC technologies [21]. Another comprehensive security framework has been presented to ensure the security of QKD protocols. It addresses finite-size security, key generation, error correction, privacy amplification, and parameter estimation. Alice and Bob exchange quantum states and then publicly communicate to reconcile their keys, ensuring minimal errors and a high level of secrecy. Parameter estimation calculates the error rates and other statistical measures to ensure the security of the key. Then, the shared keys are distilled into highly secure keys resistant to potential eavesdropping by hashing through two universal hash functions [22]. Research conducted by [20] combines QKD with logistic chaotic maps to establish a secure image transmission scheme. It utilizes the E91 QKD protocol for key distribution and logistic chaotic maps for encryption and decryption. The scheme also incorporates an eavesdropping detection mechanism based on CHSH inequality.

Cryptographic hash functions are essential for ensuring data integrity and authentication. SHA-256, part of the SHA-2 family, is widely used in various security protocols and applications. It produces a 256-bit hash value from input data, providing strong collision resistance and preimage resistance and, therefore, widely used in encryption methods. [23] integrated DNA sequence operations with a chaotic system (specifically the Lorenz system) and the Secure Hash Algorithm SHA-2 to enhance the security of image encryption. SHA-2 adds an extra layer of security and integrity to the encrypted images. Another study conducted by [24] proposed a design of a color image encryption technique performed on DNA sequences with the objective of making the design plaintext-sensitive. This scheme integrates 1D chaotic systems with the SHA-256 hash function of a plain image's RGB components to dynamically update secret keys. This process enhances security by making keys sensitive to any changes in the plain image, mitigating various plaintext attacks like chosen plaintext and known plaintext attacks. The scheme transforms real numbers into integers, converting key matrices derived from random integer sequences into DNA sequences. It employs two different diffusion mechanisms and adjusts the permutation sequence with changes in the secret key's block size. Experimental results show that the scheme is sensitive to variations in the plaintext and secret key parameters, which justifies its robustness.

Steganography [8] is the art of hidden communication over a traditional route. QKD, embedded with this security measure, introduced a new way of fortifying data. A security model visualized the idea of using the Bernstein–Vazirani algorithm for integrating data into a quantum circuit and QKD protocol BB84 to obtain a secret key, which ensures that only authorized parties can access the quantum message. This quantum steganography protocol increases the hidden channel capacity. It also withstands intercept–resend and auxiliary particle attacks due to the randomness in the distribution of information and secret messages. Also, it uses more Bell states to reduce the detectability of hidden messages [25]. [26] presented a novel protocol for quantum steganography using discrete modulation continuous variable QKD to address vulnerabilities in existing protocols. By incorporating reverse communication, the proposed protocol avoids detection by steganalysis(a procedure that focuses on detecting hidden information or data within various types of digital media), ensuring the security and secrecy of the hidden messages. Most common classical steganography technique is LSB(least significant bit) [27] based steganography, which chooses the least bit of a pixel. [28] this paper uses image steganography technique based on BES(bald eagle search) and encrypted with chaotic encryption, but there is no mention of secure key distribution. Our work comprises multi-layer security of QKD protocol with classical cryptographic encryption and hash function, all embedded together to conceal the steganographic data to a more secure extent.

2.1. Novel contributions to existing literature gap

All of these existing works have strengths and weaknesses. Chaotic encryption is secure but exhibits lower entropy values compared to AES with SHA. RSA is effective for key distribution but is vulnerable to quantum attacks that can break RSA, ECC, or any other classical key distribution scheme. LSB-based steganography is not very efficient in embedding data. While quantum and classical cryptographic methods have been extensively studied separately, their integration presents unique opportunities for enhanced security. The novelty of our approach lies in:

1. Implementation combining E91 protocol specifically with AES and SHA for steganographic security.
2. Novel integration architecture that preserves quantum security advantages while leveraging classical cryptographic efficiency.
3. Unique application of deep learning-based steganography within a quantum–classical framework.
4. Comprehensive analysis on the security metrics.

3. Proposed architecture

Our proposed architecture integrates the E91 QKD protocol with classical cryptographic methods and deep steganography to ensure a highly secure image data encryption system. This system leverages the E91 protocol for QKD to generate cryptographic keys, the Secure Hash Algorithm to hide and make fixed key lengths, AES for encryption, and deep learning models for steganography. SHA has strong collision resistance [29], gives fixed output length providing consistent key size, and has advantages over alternatives. AES has the highest security against known attacks. It is efficient for hardware/software implementation, wide industry adoption, and standardization. The architecture ensures robust data security by combining quantum and classical techniques. Fig. 1 gives a high-level overview of our Multi-Layered Security System.

3.1. Initialization

We have created a singlet circuit to create entangled qubits using Qiskit SDK [30] for quantum computing. So that they can make a secured shared key through the E91 quantum key distribution protocol, described in Appendix A.2. We added a Hadamard(H) gate to make the qubit in superposition and added a CNOT gate to control the 2nd qubit. We have explained these quantum logic gates in Appendix A.3. Fig. 2 shows the initial singlet state circuits of the E91 protocol.

3.2. Measurement

Alice and Bob select some bases independently, described in appendix 4. Then, they make independent measurements based on those bases. They communicate through classical channels to verify their same measurement bases and generate secret keys from those measurement bases. Those measurement bases are not the same for each other; they prepare those bases for the CHSH test to check the eavesdropper, which is described in Appendix A.2. Fig. 3 shows the measurement circuits used by Alice on x, $w = (\frac{(X+Z)}{\sqrt{2}})$, and z bases. Similarly, Fig. 4 shows the measurement circuits used by Bob on $w = (\frac{(X+Z)}{\sqrt{2}})$, $v = (\frac{(X-Z)}{\sqrt{2}})$, and z bases. Where x and z are pauli basis. $x = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $z = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Fig. 5 shows one of 500 combined measurement circuits of singlet state, Alice measurement, and Bob measurement circuits.

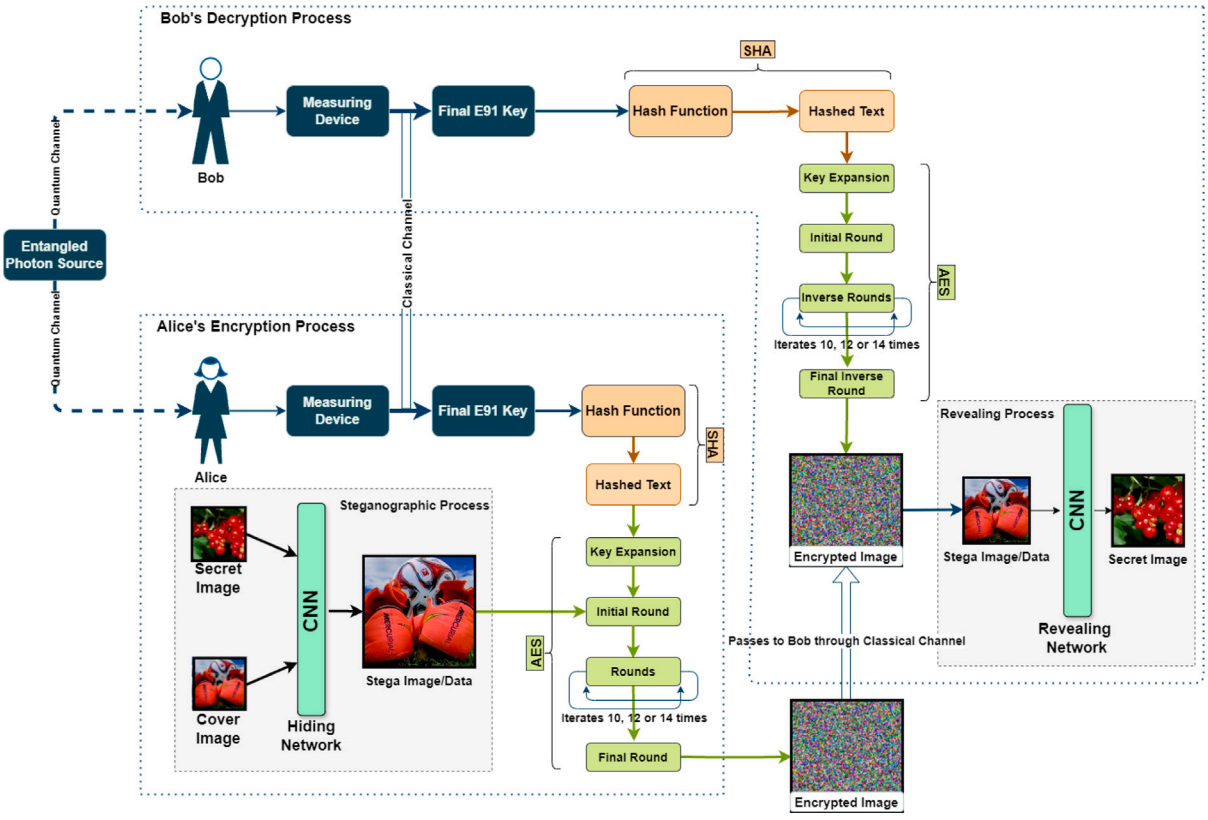


Fig. 1. A high-level overview of the proposed architecture for the Multi-Layered Security System. E91 protocol shared secret keys between Alice and Bob from an entangled photon source after doing some bases measurements described in Appendix A.2. Then, the key passes to the Secure Hash Algorithm function to create a hashed key for the secret shared key. Then, it passes to the AES encryption algorithm to encrypt the Steganographic Data, which we described in Appendix A.9. Thus, we created a robust security system to enhance the cryptographic method and steganographic security.

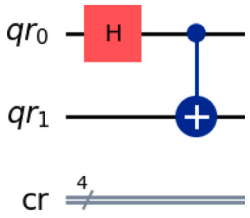


Fig. 2. Singlet state circuit to create entangled pair.

3.3. Hash function

After extracting the shared secret key from the E91 protocol, Alice and Bob create a hash text using the SHA. The detail about SHA-256 is described in Appendix A.5. Hashing helps to hide the original length of the secret key and creates a fixed length. So, if somehow an eavesdropper gets access to the hash text, they will still be unable to extract the original key.

So, we can write,

$$H = H'(E91_{key}) \tag{1}$$

Here H is the Hashed text, $H'()$ is the Hash function, and $E91_{key}$ is the shared secret key generated from the E91 protocol.

3.4. Encryption and decryption using AES

After generating the shared secret key from the E91 protocol and using the Hash function, we created a cryptographic hashed key H in Eq. (1). This hashed key H enters the Key Expansion section of AES encryption [13] and then encrypts the desired data. We have described methods related to AES in Appendix A.4. Fig. 6 shows the detailed data encryption and decryption process.

So, after encrypting data, we can write,

$$C = AES_{encrypt}(I, H'(E91_{key})) \tag{2}$$

From Eq. (1), we got, $H = H'(E91_{key})$

So, we can write,

$$C = AES_{encrypt}(I, H) \tag{3}$$

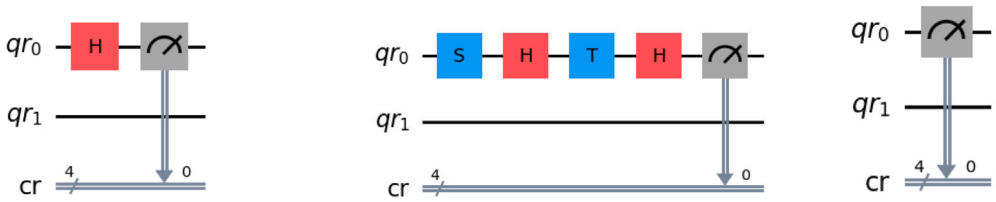
Here, C = Cipher Data, $AES_{encrypt}()$ is the encryption function, I is the stega image/plain data, H is the hashed key. Similarly, for decryption,

$$I = AES_{decrypt}(C, H) \tag{4}$$

Here, $AES_{decrypt}()$ is the decryption function.

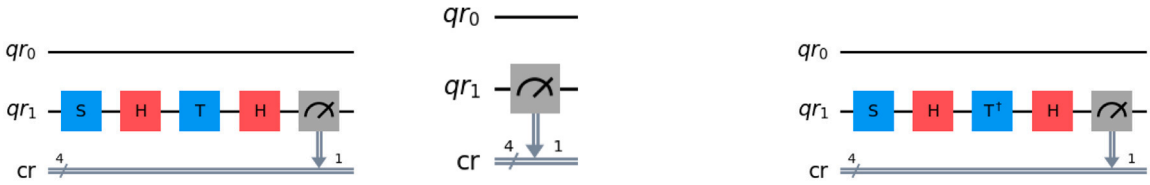
3.5. Deep learning based steganography

Our proposed framework incorporates deep learning-based steganography to enhance data security instead of traditional Least Significant Bit (LSB) techniques. The CNN-based steganography model has been derived for the GitHub repository of [31]. While LSB steganography modifies only the least significant bits of pixel values, the CNN-based approach [32] distributes the secret information across all available bits of the cover image, making the hidden data significantly more difficult to detect and extract. This deep learning methodology provides critical advantages over conventional LSB approaches [33,34]. Unlike fixed bit replacement patterns, this network dynamically learns



(a) Measurement circuit of Alice's qubit on pauli-x basis. (b) Measurement circuit of Alice's qubit on w basis. (c) Measurement circuit of Alice's qubit on pauli-z basis.

Fig. 3. Alice's measurement circuits on x, w, and z basis.



(a) Measurement circuit of Bob's qubit on w basis. (b) Measurement circuit of Bob's qubit on pauli-z basis. (c) Measurement circuit of Bob's qubit on v basis.

Fig. 4. Bob's measurement circuits on w, z, and v basis.

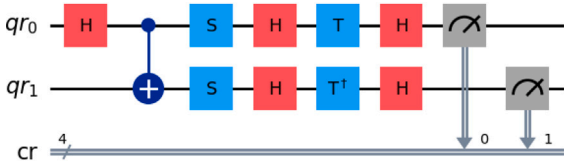


Fig. 5. One of the 500 prepared measurement circuits in different basis i.e., z, w, x, and v.

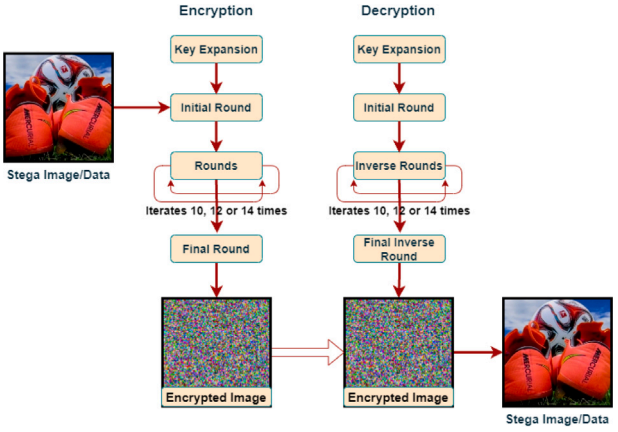


Fig. 6. An overview of AES encryption and decryption method.

optimal embedding strategies specific to the characteristics of the cover and secret images. This deep learning-based approach not only improves imperceptibility but also significantly enhances resistance to statistical steganalysis by minimizing detectable patterns [35]. In Appendix A.9, we talked about deep learning-based steganography. Fig. 7 shows how deep steganography works.

3.5.1. Data used in CNN-based steganography network

The network from [31] is trained on 1000 RGB images from the DIV2K dataset [36], with 800 images used for training, 100 images for validation, and 100 images for testing. All images were preprocessed to a standard size of 224 × 224 pixels and normalized to the [0,1] range.

3.5.2. CNN-based steganography architecture

The CNN-based steganographic system from [31] consists of two primary neural network components: (I) a hiding network and (II) a revealing network.

Hiding Network: The Hiding Network accepts both the secret and cover images as inputs and combines them into a single stega image that appears visually similar to the original cover image. [31] employs a multi-scale architecture with three parallel branches, each utilizing different convolution kernel sizes ($3 \times 3, 4 \times 4$, and 5×5) to capture features at varying scales. Each branch processes the inputs through four consecutive convolutional layers with 50 filters, using ReLU activation functions to introduce non-linearity. The features from all branches are concatenated and further processed through a final 1×1 convolutional layer to produce the stega image. This network ensures efficient and robust embedding of the secret image into the carrier.

Reveal Network: The Reveal Network from [31] mirrors the architecture of the Hiding Network but focuses on extracting and reconstructing the hidden secret image from the stega image. It begins with a Gaussian noise layer ($\sigma = 0.1$) at the input to simulate transmission noise and enhance robustness against minor distortions or perturbations. The network employs the same multi-scale design with parallel branches using $3 \times 3, 4 \times 4$, and 5×5 convolutional layers to extract features, followed by concatenation of these features. The outputs are processed through additional convolutional layers to decode and reconstruct the secret image, ensuring effective recovery even under noisy conditions.

4. Result analysis

We have used Qiskit SDK for Quantum Computing by IBM to generate quantum key distribution. Jupyter Notebook and Python were

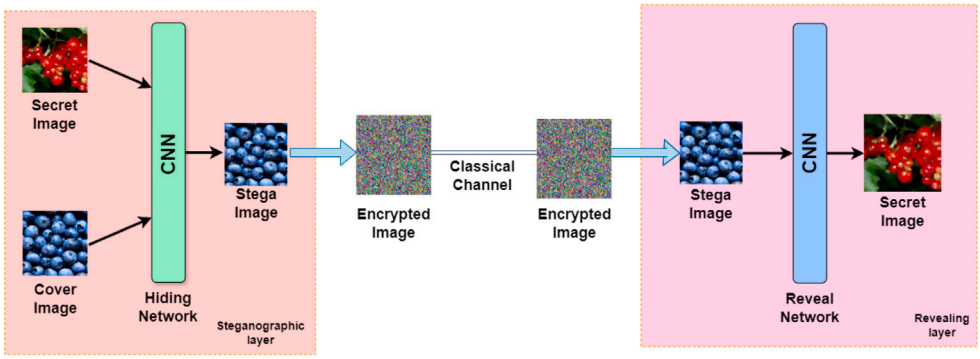


Fig. 7. Overview of the CNN-based steganography system used in [31]. The Hiding Network embeds a secret image into a cover image using a multi-scale architecture with parallel convolutional branches (3×3, 4×4, 5×5 kernels), producing a stega image. The stega image is encrypted, transmitted through a classical channel, and decrypted. The Reveal Network, mirroring the Hiding Network, reconstructs the secret image with a Gaussian noise layer ($\sigma = 0.1$).

Table 1
Entrophy computation of encrypted images of different pixel sizes.

Pixel size	Entropy cipher
64 × 64	7.9842
128 × 128	7.9894
256 × 256	7.9986
512 × 512	7.9995
Average	**7.9929**

Table 2
Key generation rate.

Singlet state	Key length	Key generation time (s)	Key generation rate (bps)
25	7	4.39	1.59
100	25	4.66	5.37
250	57	5.42	10.52
500	106	8.66	12.24
Average		**5.78**	**7.43**

Table 3
Execution time complexity of the proposed system.

Pixel Size	Encryption time (s)	Decryption time (s)
64 × 64	0.00098	0.00100
128 × 128	0.00100	0.00001
256 × 256	0.00100	0.00200
512 × 512	0.00300	0.00399
Average	**0.00149**	**0.00175**

used to get different results like key generation, encryption rate, entropy analysis and differential attack analysis. All the simulation and performance evaluations of the proposed system were performed on AMD Ryzen 5 3550H with Radeon Vega Mobile Gfx 2.10 GHz, 16 GB RAM, NVIDIA GeForce GTX 1050 with 64-bit Windows 11. We have also shown different histogram results of an encryption and decryption image.

4.1. Entropy analysis of encrypted images

The Shannon entropy [37] of a cipher is a measure of its randomness and unpredictability, which are crucial properties for secure encryption. High entropy means that the encrypted data appears random and has no discernible patterns, making it difficult for attackers to predict or deduce the original data.

$$H(x) = -\sum_{i=1}^{n} P(x_i) log_b(P(x_i)) \tag{5}$$

Here, $H(x)$ is the entropy of the random variable x. $P(x_i)$ is the probability outcome. An encryption technique should ideally output an encrypted image with an entropy value of 8, computed using Eq. (5). Table 1 shows the obtained entropy values for encrypted images with different pixel sizes.

The results show that, regardless of image resolution, entropy levels constantly approach the ideal value of 8 with minimal variance. The entropy values range from 7.9842 for a 64 × 64 pixel image to 7.9995 for a 512 × 512 pixel image, with an average of 7.9929 for all examined images.

4.2. Key generation rate of E91 protocol

Here, we generated the key generation rate and time for the E91 protocol using Qiskit SDK for Quantum Computing. Table 2 shows the different rates and times in seconds and bits per second. The result reveals that key generation performance is influenced by the size of the singlet state. As the number of singlet states increases, so does the key length and generation time, resulting in varying key generation rates. For instance, a 7-bit key was generated in 4.39 s with a singlet state of 25, while a 106-bit key was generated in 8.66 s with a singlet state of 500.

4.3. Encryption and decryption rate

Here, we will see the encryption and decryption [38] time rate of our proposed system for different image pixel sizes. Table 3 shows the results of execution time complexity. The results show that the encryption and decryption times increase slightly as the pixel size of the image grows. For instance, while a 64 × 64 pixel image requires only 0.00098 s for encryption and 0.00100 s for decryption, a larger 512 × 512 pixel image takes 0.00300 s to encrypt and 0.00399 s to decrypt. The average encryption time across all tested pixel sizes is 0.00149 s, with an average decryption time of 0.00175 s.

4.4. Inference time analysis

To evaluate the practical applicability of our system, we conducted a comprehensive analysis of inference times for both the hiding and revealing networks across different images. Table 4 presents the average inference times measured across multiple runs. The hiding network shows slightly longer inference times compared to the revealing network, which is expected given its more complex architecture incorporating both the cover and secret image processing. The average total processing time of 8.51758 s across all images.

Table 4
Inference time analysis for different images.

Images	Hide time (s)	Reveal time (s)	Total time (s)
Image 1	6.58932	2.44001	9.02933
Image 2	5.84816	2.33116	8.17932
Image 3	6.32971	2.26387	8.59358
Image 4	6.05916	2.55339	8.61255
Image 5	5.83294	2.34020	8.17314
Average	**6.13186**	**2.38573**	**8.51758**

4.5. Histogram analysis

In Fig. 8 we have shown the different histograms [39] of different pixel images. The histogram analysis demonstrates that the encryption process successfully converted the original image into an unrecognizable form by randomizing pixel intensities, as evidenced by the uniform histogram of the encrypted image. The decrypted image's histogram, with its non-uniform distribution, indicates a successful restoration of the original image's pixel intensity structure. This confirms that the encryption and decryption processes worked as intended, ensuring both security and data integrity.

4.6. Analysis of key generation for our multi-layered security system

Table 5 shows different types of analysis of our proposed multi-layered security system, which includes the number of singlet states, shared keys, the hashed value, and the hash length.

The analysis shows that the length of the hash key remains constant regardless of how many singlet states are employed to construct the binary key. This is because the hashing technique generates a fixed-length output, which is common among cryptographic hash functions.

4.7. System performance analysis

To evaluate the practical applicability of our system, we conducted a comprehensive timing analysis across different stages of the pipeline. Table 6 presents the average processing times across all tested image resolutions (64 × 64 to 512 × 512 pixels). The results demonstrate that our multi-layered security system maintains efficient processing times across all components. The steganographic operations (hiding and revealing) constitute the majority of the processing time, the key generation time of QKD E91 averaged 5.78 s, while the encryption and decryption operations add minimal overhead (averaging just 0.00324 s combined). The total system processing time of 14.30083 s indicates suitability for real-time applications.

4.8. Key-sensitivity analysis

Key sensitivity is an essential property of cryptographic systems, ensuring that even a minor change in the key produces a vastly different ciphertext. This property guarantees that an attacker cannot derive useful information from closely related keys.

To demonstrate key sensitivity in our system, we conducted an experiment where we encrypted some color images with K1 and tried to decrypt the images with K2(where we altered one bit). The resulting ciphertexts were then compared to evaluate the difference.

For instance, using the keys, $K1 = 1010011110001000111011$ and $K2 = 0010011110001000111011$, the corresponding Hashed value of $K1 = dd116ea845b69b000cfde2831b67e5ac53544fd9688b86123fef6235e34af651$ and the corresponding Hashed value of $K2 = 9a750c1f027d9db0e32063770813af1fc07108d7bc19c96e46c41f88379d684b$.

Figs. 9 and 10 show how highly sensitive our encryption system is. This ensures that even if an attacker gets close to the correct key, the decryption will still fail completely, thereby maintaining the security of the system.

4.9. NPCR and UACI attack analysis

NPCR [40] (Number of Pixels Change Rate) and UACI [41] (Unified Average Changing Intensity) are metrics used to evaluate the effectiveness of image encryption algorithms by measuring the difference between an original image and its encrypted version. NPCR measures the percentage of different pixel values between the original and encrypted images. It indicates how many pixels have changed due to the encryption process. The formula for NPCR is:

$$NPCR = \frac{\sum_{i,j} D(i,j)}{M \times N} \times 100\% \tag{6}$$

where $D(i,j)$ is a binary array where $D(i,j) = 1$ if the pixel value at position (i,j) in the original image is different from the encrypted image, and $D(i,j) = 0$ otherwise. M and N are the dimensions of the image.

UACI measures the average intensity of differences between the original and encrypted images. It quantifies the average intensity of the differences in pixel values. The formula for UACI is:

$$UACI = \frac{1}{M \times N} \sum_{i,j} \frac{|C_1(i,j) - C_2(i,j)|}{255} \times 100\% \tag{7}$$

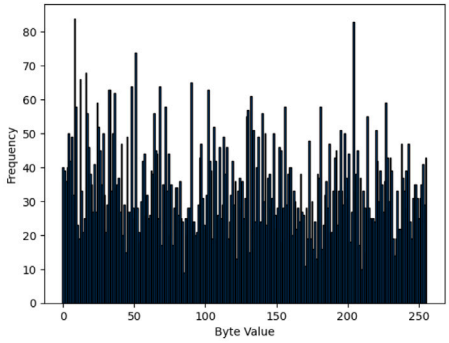
where, $C_1(i,j)$ and $C_2(i,j)$ are the pixel values at position (i,j) in the original and encrypted images, respectively and $D(i,j) = 0$ otherwise. M and N are the dimensions of the image. Table 7 shows differential attack analysis. The results show that the encryption technique effectively disperses pixel changes, making it resistant to differential attacks. The NCPR values, which measure the proportion of pixels that change when an input image is slightly adjusted, are consistently high at 99.6928% across all tested pixel sizes. The UACI values, which measure the average intensity difference between original and encrypted images, show strong performance at 56.1549%, indicating that even small changes in input cause significant changes in the encrypted image.

4.10. Correlation analysis

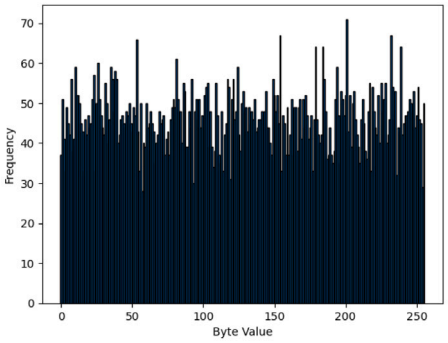
A comprehensive correlation analysis was performed to evaluate the security of our proposed encryption scheme by examining pixel relationships in the horizontal, vertical, and diagonal directions. The correlation coefficients were calculated between adjacent pixels for the original image, steganographic image, and the encrypted version. Figs. 11 and 12 shows the correlation analysis for our proposed system. The original unprocessed image exhibits strong pixel correlations as expected for natural images, with correlation coefficients of 0.9246 (horizontal), 0.8794 (vertical), and 0.8370 (diagonal). These high values indicate strong linear relationships between adjacent pixels in all directions. After applying our steganographic technique, the correlation values remain similar at 0.9381 (horizontal), 0.8957 (vertical), and 0.8541 (diagonal), demonstrating the steganography's ability to preserve the natural image characteristics while hiding information. However, after encryption, the correlation coefficients drop dramatically to 0.1137 (horizontal), 0.1007 (vertical), and 0.0925 (diagonal). These values approaching zero indicate that our encryption scheme successfully destroys the inherent pixel relationships in all directions. The correlation graphs visually confirm this transformation — while the original and steganographic images show clear linear patterns along the diagonal, the encrypted image exhibits a uniform, noise-like distribution across the entire pixel value range.

4.11. Comparative analysis with previous methods

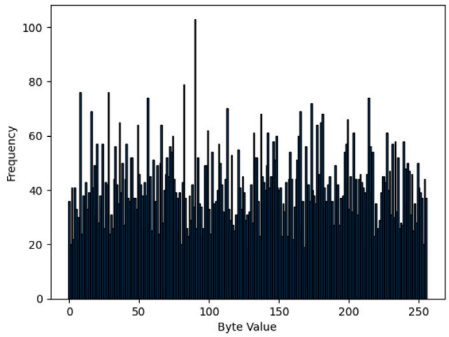
To evaluate the effectiveness of our multi-layered security approach, we compare our results with several existing methods across key security metrics. Table 8 presents a comprehensive comparison of entropy values, NPCR, UACI, and execution times across different approaches. The comparative analysis demonstrates that our proposed system outperforms other systems across most key metrics. Notably, it surpasses all existing hybrid quantum–classical cryptographic techniques, offering superior performance.



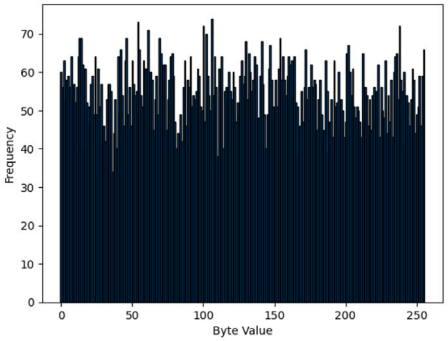
(a) Original Image(64×64)



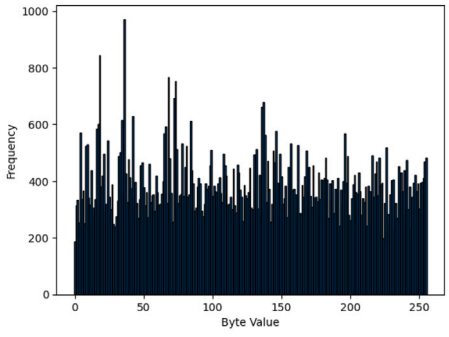
(b) Encrypted Image(64×64)



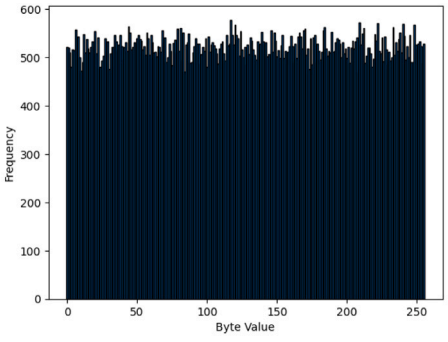
(c) Original Image(128×128).



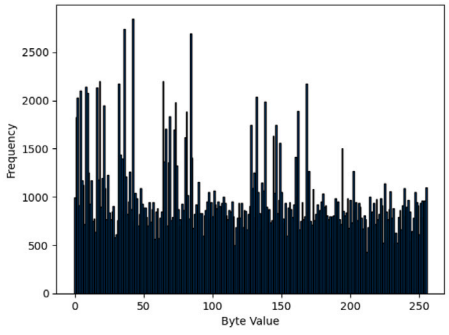
(d) Encrypted Image(128×128).



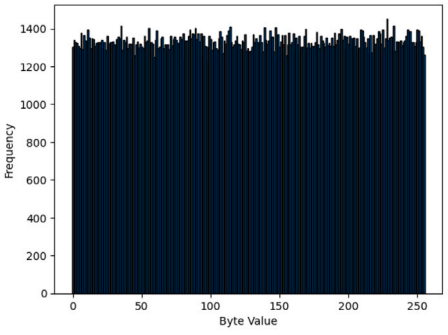
(e) Original Image(256×256)



(f) Encrypted Image(256×256).



(g) Encrypted Image(512×512).



(h) Encrypted Image(512×512).

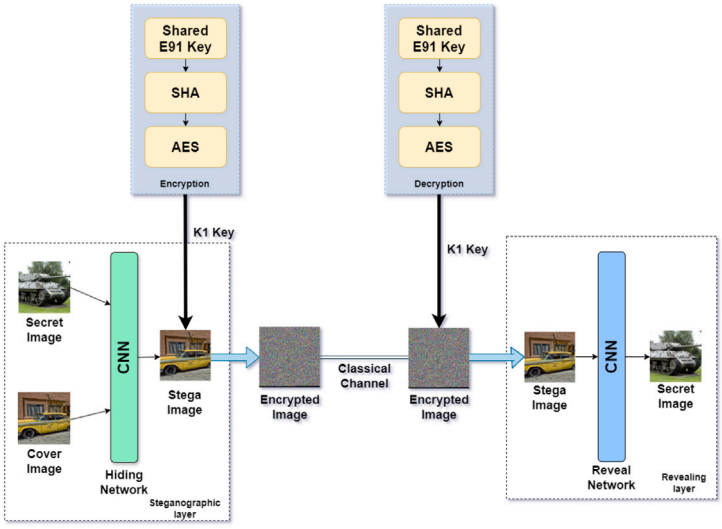
Fig. 8. Histogram analysis of the images.

5. Discussion and findings

In our research, we have employed the Quantum Key Distribution Protocol E91 alongside some classical cryptographic methods. We encrypted and decrypted color images of various pixel densities, specifically 64×64, 128×128, 256×256, and 512×512. Our encryption process was independent of the resolution of the images. We calculated and analyzed various results. We observed that the higher the pixel density of the images we encrypted, the more secure the encryption became. We calculated the entropy of the images and obtained remarkable results, with values ≈ 8 on an 8-point scale. We also analyzed the differential attacks, such as NPCR and UACI, and found near-perfect results, as detailed in Section 4. Furthermore, we determined the key generation rate and key length for different numbers

Table 5
Number of singlet states, quantum key generation, hash value and hash length.

Number of singlet states	Key generated (binary) with varying key length	Binary key length	Generated hashed key (Hex) with fixed key length	Hashed key length
25	10 010 111	8	459c2daec5458568864215c57d12fa0a e28243b080971bb90d09fe020f8f265e	64
100	1010011110001000111011	22	dd116ea845b69b000cfde2831b67e5ac 53544fd9688b86123fef6235e34af651	64
250	0110110001111001010101101110000101000011101100100	49	6c26412c58131663b4034e37bf71318c 115589186085671c27f25b368e4cbb4e	64
500	0100010010101000101011100101000100001001111001000111001101001101100111100011111101000101100000000	98	b0c1de302023c06df2ca56b7206da959 dccc9cc8d25f983f921b51664a7e2851	64



(a) Full multi-layered security system, Encrypted and Decrypted Image using K1 key



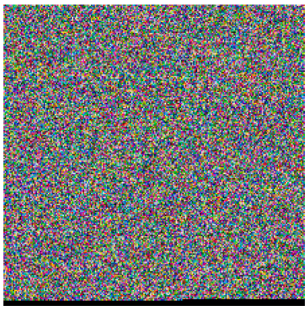
(b) Secret image



(c) Cover image



(d) Stenographic image



(e) Encrypted image

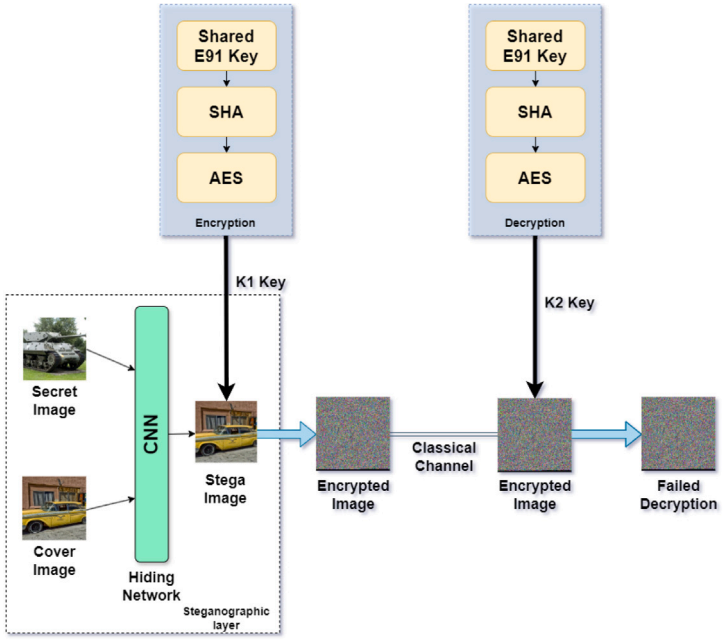


(f) Decrypted image



(g) Retrived secret image

Fig. 9. Transmission process of the secret image using K1 key.



(a) Full multi-layered security system, Encrypted using K1 key and Decrypted using K2 key



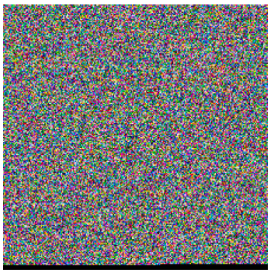
(b) Secret image



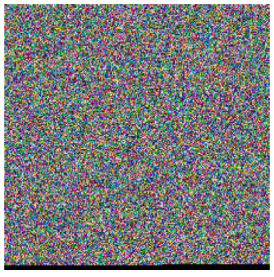
(c) Cover image



(d) Stenographic image



(e) Encrypted image



(f) Failed Decrypted image

Fig. 10. Transmission process of the secret image using K1 and K2 key.

Table 6
Average system timing analysis.

Operation	Average time (s)
Key generation	5.78000
Hide network	6.13186
Reveal network	2.38573
Encryption	0.00149
Decryption	0.00175
Total system time	**14.30083**

Table 7
Computations of differential attack analysis.

Pixel size	NCPR (%)	UACI (%)
64 × 64	99.6337	54.4415
128 × 128	99.8229	57.7351
256 × 256	99.5432	55.4843
512 × 512	99.7717	56.9590
Average	**99.6928**	**56.1549**

of singlet states. Our findings indicate that around 25% of the singlet states are used to generate the key, while the remaining singlet states are utilized to calculate the CHSH inequality, ensuring the detection of any eavesdroppers. We also measured the key generation time and rate, as well as the encryption and decryption times of our proposed system. As expected, larger pixel sizes required more time for encryption. We created histogram images to visually compare the differences between

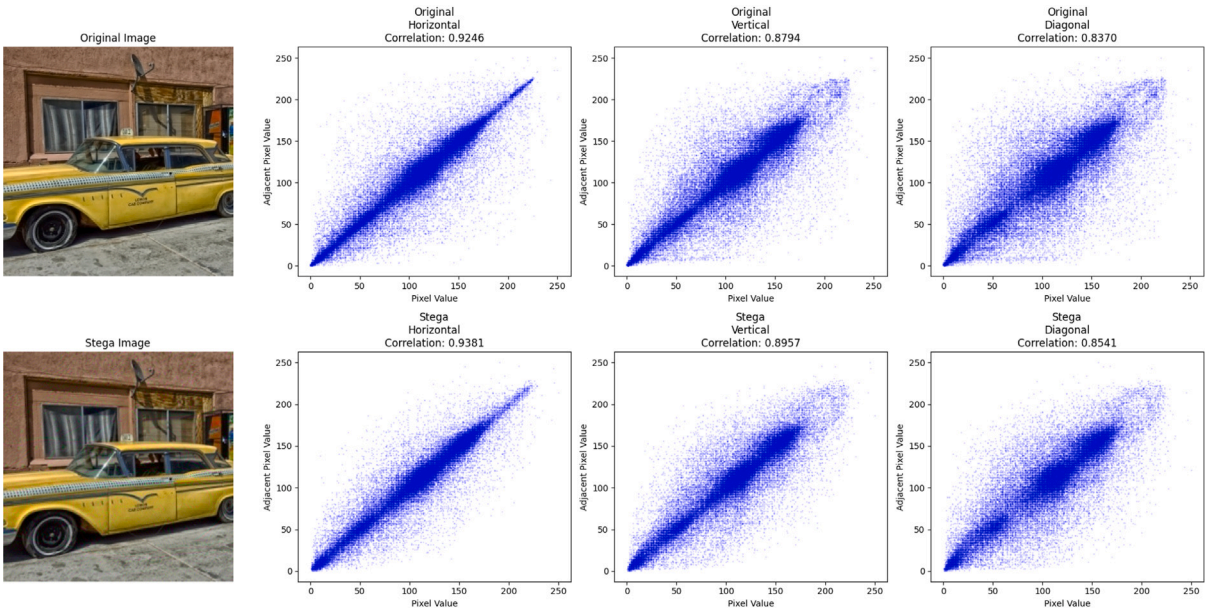


Fig. 11. Correlation analysis of adjacent pixels in horizontal, vertical, and diagonal directions for original and steganographic images.

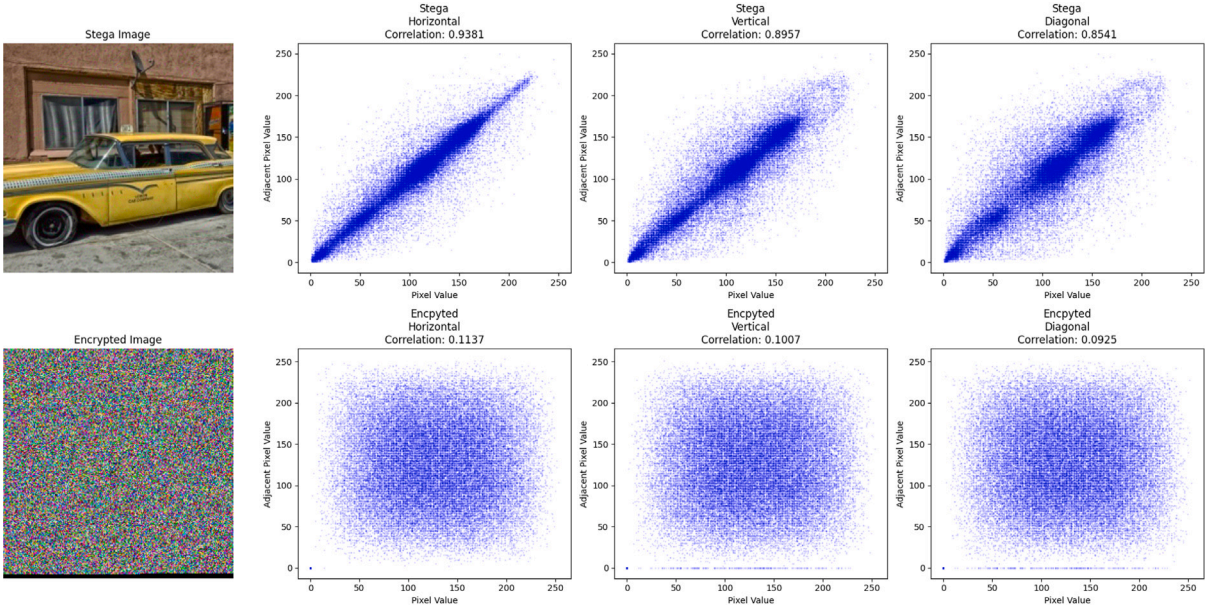


Fig. 12. Correlation analysis of adjacent pixels in horizontal, vertical, and diagonal directions for original and encrypted images.

Table 8
Comparison with existing methods.

Method	Avg. Entropy	NCPR (%)	UACI (%)	Avg. Encryption (s)	Hybrid architecture type
Proposed method	**7.9929**	**99.6928**	**56.1549**	$\mathbf{1.49 \times 10^{-3}}$	**QKD, Steganography, AES, SHA**
S. Priya et al. [42]	7.9865	99.5416	33.4440	N/A	ECC, Steganography, Chaos
T. Janani et al. [43]	7.9782	99.7172	33.4511	8.5×10^{-1}	QKD, Quantum Image Encryption
R. Rahman et al. [20]	7.3500	99.6600	34.6200	7.481×10^{-8}	QKD, Chaos
M. Naim et al. [44]	7.9993	99.6033	33.4588	N/A	Chaos and AES
S. Kumar et al. [45]	7.9983	99.5740	33.3900	858.87×10^{-3}	ECC, Chaos
H. Liu et al. [46]	7.9983	99.6065	33.4460	9.28 (MB/s)	OTP, Chaos

the encrypted and original images. We also observed the key sensitivity of our proposed system and conducted a thorough key analysis of our multi-layered security system. Implementing quantum components is a very challenging task to this date. We need an entangled photon source that will create entanglement between two photons that have to transmit through a quantum channel, which can be created using fiber optics or free-space optical communication channels. We need high-efficiency single-photon detectors (e.g., avalanche photo-diodes

or superconducting nanowire detectors) for measuring quantum states. Finally, precise timing synchronization devices are used to correlate measurements between parties.

5.1. Limitations of proposed method

While our hybrid system offers enhanced security by combining the quantum-resistant properties of QKD with the efficiency of AES and SHA, it is important to acknowledge the following limitations:

1. QKD systems are currently limited by transmission distances due to photon loss in optical fibers.
2. QKD systems require specialized hardware, making them more complex and expensive to deploy and maintain compared to classical encryption systems.
3. Ensuring precise synchronization in the real world between the QKD, AES and SHA components is crucial for secure operation.
4. The quantum key distribution process introduces additional latency, which may be significant for real-time applications.

By carefully considering these limitations and exploring potential solutions, we can maximize the benefits of our hybrid system.

6. Conclusion and future scope

In our research, we proposed a robust and secure image data encryption system that integrates QKD using the E91 protocol with classical cryptographic techniques, specifically the Advanced Encryption Standard (AES) and Secure Hash Algorithm. The system capitalizes on the inherent security advantages of quantum mechanics to ensure the confidentiality and integrity of transmitted data. Key contributions of this work include the utilization of the E91 protocol for generating cryptographic keys based on entangled photon pairs, the integration of QKD with AES to provide a strong encryption mechanism for securing image data, and the employment of deep learning-based steganography to embed encrypted images within cover images, further enhancing the security and privacy of data transmission. This method allows for the hidden transmission of sensitive information, making it less susceptible to detection. Our system demonstrates a practical application of combining quantum and classical cryptographic techniques with steganography, resulting in a highly secure image data encryption system. The proposed architecture is not only robust against potential eavesdropping but also ensures that the transmitted data remains confidential and intact. While the suggested approach represents substantial advances in secure image data encryption, there are various opportunities for further research and enhancement. Our future work may include, (I) Exploring hybrid quantum–classical systems that combine the strengths of quantum and classical cryptography in more complex and diverse applications could lead to the development of protocols that seamlessly integrate with existing classical infrastructure. (II) In a post-quantum era, if a system's security is compromised, a post-quantum algorithm can be seamlessly integrated into any layer of our hybrid architecture. This demonstrates the architecture's adaptability and robustness in the face of evolving security threats. (III) Exploring scalable QKD solutions and lighter protocols, such as BB84 and B92, for large networks and longer communication distances while assessing their feasibility within hybrid architectures. (IV) Investigate quantum error correction techniques to improve the reliability of QKD in noisy environments and enhance key generation over long distances. (V) Enhancing the current deep learning-based steganography techniques by improving neural network architectures and training methods to strengthen the system's resistance to data extraction. (VI) Improving practical implementation feasibility through developing more efficient key management systems for operational simplicity, creating standardized interfaces for seamless quantum–classical integration. (VII) A dedicated and comprehensive study to evaluate hybrid quantum–classical frameworks' practicality, scalability, and security implications in diverse use cases. Our work not only addresses current security challenges but also lays the groundwork for future developments in secure communication technologies, marking a step forward in the era of hybrid quantum-enhanced cryptography. The proposed future directions aim to overcome current limitations while enhancing the system's practicality and scalability for widespread adoption.

CRediT authorship contribution statement

Arman Sykot: Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Md Shawmoon Azad:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Wahida Rahman Tanha:** Writing – review & editing, Writing – original draft, Validation, Software, Formal analysis. **B.M. Monjur Morshed:** Software, Methodology. **Syed Emad Uddin Shubha:** Validation, Supervision, Methodology. **M.R.C. Mahdy:** Validation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Quantum states

In quantum mechanics, a quantum state [47] describes the state of a quantum system. For a qubit, has a two-dimensional state space, which can be expressed as a linear combination (or superposition) of the basis states $|0\rangle$ and $|1\rangle$. Any arbitrary state of a qubit can be represented as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{8}$$

Here, α and β are complex numbers. For the state $|\psi\rangle$ to be normalized (i.e., to be a unit vector), it must satisfy the condition:

$$\langle\psi|\psi\rangle = 1 \tag{9}$$

This normalization condition implies:

$$|\alpha|^2 + |\beta|^2 = 1 \tag{10}$$

This ensures that the total probability of finding the qubit in either state $|0\rangle$ or $|1\rangle$ is 1

A.1. Entanglement

When two qubits are entangled [47], their joint state cannot be expressed as a simple product of individual states. Instead, the state of the system is described by a single vector in the combined Hilbert space of the two qubits. So, a pure bipartite state $|\Psi\rangle_{AB}$ is entangled if it cannot be written as a product state $|\psi\rangle_A \otimes |\phi\rangle_B$ for any choices of states $|\psi\rangle_A$ and $|\phi\rangle_B$.

If we consider a density matrix of a pure state is ρ, where $\rho = |\psi\rangle\langle\psi|$.

A system is entangled if its density matrix cannot be written as a product of the density matrices of its subsystems.

Mathematically, this is expressed as:

$$\rho \neq \rho_A \otimes \rho_B \tag{11}$$

where ρ_A and ρ_B are the density matrices of the subsystems A and B, respectively.

So, Bell states are defined as:

$$\psi_{\pm} = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle) \tag{12}$$

$$\phi_{\pm} = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle) \tag{13}$$

A.2. E91 protocol

1. **Introduction:**[6] In 1991, Artur Ekert developed a new approach to QKD by introducing the E91 protocol. This protocol ensures security through a Bell-like test to detect eavesdroppers (Eve).
2. **Entangled Particle Pairs:** A single source emits pairs of entangled particles, typically polarized photons. Each pair is described by a Bell state, specifically the singlet state:
   $$\psi_{\pm} = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle) \tag{14}$$
   This state implies that if one particle is measured to be in state $|0\rangle$, the other will be in state $|1\rangle$, and vice versa. The entangled particles are separated, with one particle sent to Alice and the other to Bob.
3. **Measurement Process:**[48] Alice and Bob each measure the polarization of their received photons. They randomly choose from three possible measurement bases, aligned according to the Clauser, Horne, Shimony, and Holt (CHSH) test.
4. **Measurement bases:**
   Alice's bases are:
   $$a_1 = 0, a_2 = \frac{\pi}{4}, a_3 = \frac{\pi}{2} \tag{15}$$
   corresponding to the bases Z, $\frac{(X+Z)}{\sqrt{2}}$, and X, respectively.
   Bob's bases are:
   $$b_1 = \frac{\pi}{4}, b_2 = \frac{\pi}{2}, b_3 = \frac{3\pi}{2} \tag{16}$$
   corresponding to the bases $\frac{(X+Z)}{\sqrt{2}}$, X, and $\frac{(X-Z)}{\sqrt{2}}$, respectively.
5. **Basis Discussion and Eavesdropper Detection:** Alice and Bob publicly discuss which measurement bases they used without revealing their measurement outcomes. They use instances where they choose different bases to detect the presence of an eavesdropper.
6. **CHSH Test:** The CHSH inequality helps detect eavesdropping by checking the correlation between Alice's and Bob's measurements:
   $$E = \langle a_1 b_1\rangle - \langle a_1 b_3\rangle + \langle a_3 b_1\rangle + \langle a_3 b_3\rangle \tag{17}$$
   $\langle a_i b_j\rangle$ represents the expectation value when Alice measures using basis a_i and Bob using basis b_j. If $-2 \le E \le 2$, it indicates potential eavesdropping or issues with the measurement devices. In an ideal, eavesdropper-free system, the expected value is $-2\sqrt{2}$, indicating maximal violation of the CHSH inequality.
7. **State of Entangled Photons:** Entangled photons passing through a depolarizing channel can be described by the isotropic mixed state [47]:
   $$\rho_{\psi} = p|\psi\rangle\langle\psi| + \frac{(1-p)}{4} I \tag{18}$$
   Here, p is the probability of maintaining the pure entangled state, and I is the identity matrix. The CHSH test shows maximal violation $-2\sqrt{2}$ when $p = 1$, meaning the channel is free from eavesdropping.
8. **Key Generation:** When Alice and Bob detect maximal violation of the CHSH test, they are confident their data is secure and not influenced by an eavesdropper. They use the instances where they chose the same measurement bases to generate a shared private key, utilizing their perfectly anti-correlated measurement results.
9. **Essential Features of E91:** The E91 protocol capitalizes on the non-local nature of entangled states, which ensures security against eavesdropping. Eve's intervention would introduce detectable changes in the entanglement correlations, affecting the non-locality and alerting Alice and Bob to her presence. This protocol, therefore, provides a robust framework for secure quantum communication.

A.3. Quantum logic gates

The logic flow in a quantum circuit [49] involves the arrival of an input qubit state (which is a quantum state) that is operated upon by a unitary operator, and an output qubit state is obtained, following the laws or postulates of quantum mechanics.

If U is a unitary operator representing a quantum gate, and $|\psi\rangle$ is the input qubit state, the output qubit state $|\psi'\rangle$ is given by:

$$|\psi'\rangle = U|\psi\rangle \tag{19}$$

Unitary operators have the property that satisfies $U^{\dagger}U = UU^{\dagger} = I$, where $U^{\dagger}$ is the Hermitian Conjugate of U, and I is the identity matrix. This property ensures that quantum gates are reversible and preserve the total probability.

Some important quantum logic gates are given below which we need to implement the QKD protocol.

1. **Pauli-X Gate:** The Pauli-X gate flips the state of a qubit, analogous to the classical NOT gate.
   It is represented as:
   $$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{20}$$
2. **Hadamard Gate:** The Hadamard gate creates superposition states from basis states. It is essential in algorithms that rely on quantum parallelism.
   The Hadamard gate is represented as:
   $$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{21}$$
3. **Controlled-NOT (C-NOT) Gate:** The C-NOT gate is a two-qubit gate that flips the state of the target qubit if the control qubit is in the state $|1\rangle$.
   It is represented as:
   $$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{22}$$
4. **Phase Gate:** The Phase gate (S gate) introduces a phase shift of $\frac{\pi}{2}$.
   It is represented as:
   $$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \tag{23}$$
5. **T Gate:** The T gate, or $\frac{\pi}{8}$ gate, introduces a phase shift of $\frac{\pi}{4}$.
   It is represented as:
   $$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{4}} \end{pmatrix} \tag{24}$$

A.4. Advanced Encryption Standard (AES)

The Advanced Encryption Standard (AES) is a symmetric encryption algorithm widely used across various applications for securing sensitive data. Developed by Joan Daemen and Vincent Rijmen, AES was adopted as a standard by the National Institute of Standards and Technology (NIST) in 2001. It is known for its efficiency, security, and flexibility.

AES utilizes a design principle called a substitution-permutation network, making it efficient for both software and hardware implementations. It is a specific variant of the Rijndael algorithm, featuring a fixed block size of 128 bits and key sizes of 128, 192, or 256 bits. In comparison, the Rijndael algorithm itself allows for block and key sizes that can be any multiple of 32 bits, ranging from 128 to 256 bits.

Description [13] of the AES algorithm:

1. KeyExpansion: Round keys are derived from the cipher key using the AES key schedule. AES requires a separate 128-bit round key block for each round plus one more.
2. Initial round key addition:
   (a) AddRoundKey — each byte of the state is combined with a byte of the round key using bitwise xor.
3. 9, 11 or 13 rounds:
   (a) SubBytes — A non-linear substitution step where each byte is replaced with another according to Rijndael S-box.
   (b) ShiftRows — A transposition step where the last three rows of the state are shifted cyclically over a certain number of steps.
   (c) MixColumns – A linear mixing operation that operates on the columns of the state, combining the four bytes in each column.
   (d) AddRoundKey
4. Final round (making 10, 12 or 14 rounds in total):
   (a) SubBytes
   (b) ShiftRows
   (c) AddRoundKey

SubBytes() Transformation: In the SubBytes step, each byte $a_{i,j}$ in the state array is replaced by a SubByte $S(a_{i,j})$ using an 8-bit substitution box. This S-box, which is invertible, is constructed by composing two transformations:

1. Take the multiplicative inverse in the finite field $GF(2^8)$; the element {00} is mapped to itself.
2. Apply the following affine transformation (over $GF(2)$):

$$b_i = b_i \oplus b_{(i+4)mod8} \oplus b_{(i+5)mod8} \oplus b_{(i+6)mod8} \oplus b_{(i+7)mod8} \oplus c_i \quad (25)$$

for $0 \le i < 8$, where b_i is the ith bit of the byte, and c_i is the ith bit of a byte c with the value {63} or {01100011}. Here and elsewhere, a prime on a variable (e.g., b') indicates that the variable is to be updated with the value on the right.

ShiftRows() Transformation: In the ShiftRows() transformation, the bytes in the last three rows of the State are cyclically shifted over different numbers of bytes (offsets). The first row, $r = 0$, is not shifted.

Specifically, the ShiftRows() transformation proceeds as follows:

$$S_{r,c} = S_{r,(c+shift(r,Nb))mod\,Nb} \ for\ 0 < r < 4\ and\ 0 \le c < Nb \quad (26)$$

where the shift value $shift(r, Nb)$ depends on the row number, r, as follows (Number of Columns, for this standard $Nb = 4$):

$$shift(1,4) = 1;\ shift(2,4) = 2;\ shift(3,4) = 3 \quad (27)$$

MixColumns() Transformation: The MixColumns() transformation operates on the State column-by-column, treating each column as a four-term polynomial. The columns are considered as polynomials over $GF(2^8)$ and multiplied modulo x^4+1 with a fixed polynomial $a(x)$, given by

$$a(x) = \{03\}x^3 + \{01\}x^2 + \{01\}x + \{02\} \quad (28)$$

This can be written as a matrix multiplication.

$$s'(x) = a(x) \oplus s(x) : \begin{bmatrix} s'_{0,c} \\ s'_{1,c} \\ s'_{2,c} \\ s'_{3,c} \end{bmatrix} = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} s_{0,c} \\ s_{1,c} \\ s_{2,c} \\ s_{3,c} \end{bmatrix} \quad (29)$$

AddRoundKey() Transformation: In the AddRoundKey() transformation, a Round Key is added to the State by a simple bitwise XOR operation. Each Round Key consists of Nb words from the key schedule. Those Nb words are each added into the columns of the State, such that

$$\left[s'_{0,c}, s'_{1,c}, s'_{2,c}, s'_{3,c}\right] = \left[s'_{0,c}, s'_{1,c}, s'_{2,c}, s'_{3,c}\right] \oplus \left[W_{round*Nb+c}\right] \quad (30)$$

where $[w_i]$ is the key schedule words, and round is a value in the range $0 \le round \le Nr$ ($Number\ of\ rounds = 10, 12, 14$).

A.5. Secure Hash Algorithm (SHA)

The Secure Hash Algorithm (SHA) [50] family is a set of cryptographic hash functions designed to provide strong data integrity and security. SHA was developed by the NIST and has various versions, including SHA-1, SHA-2, and SHA-3. Each version of SHA processes [51] data in fixed-size blocks and produces a fixed-size hash value, making it ideal for various security applications such as digital signatures, message integrity verification, and password hashing.

A.6. Steps of SHA-256

Here, we will focus on the SHA-2 family, specifically SHA-256, which is widely used due to its balance of security and performance. SHA-256 operates through several main steps: message padding, parsing, initialization, message scheduling, and the compression function.

1. **Message Padding:** The message to be hashed is padded to ensure its length is a multiple of 512 bits (64 bytes). Padding is done as follows:
   (a) Append a '1' bit: Add a single '1' bit to the end of the message.
   (b) Append '0' bits: Add '0' bits until the length of the message is 64 bits less than a multiple of 512.
   (c) Append the length: Append the 64-bit representation of the original message length (in bits) to the end of the padded message.
2. **Parsing the Message:** The padded message is divided into 512-bit (64-byte) blocks. Each block is further divided into sixteen 32-bit words:

$$M = M_1, M_2, M3, \ldots .., M_N \quad (31)$$

3. **Initialization of Hash Values:** SHA-256 uses eight initial hash values, each 32 bits in length. These values are derived from the fractional parts of the square roots of the first eight prime numbers:

$$\begin{aligned} &H_0 = 0x6a09e667, H_1 = 0xbb67ae85, \\ &H_2 = 0x3c6ef372, H_3 = 0xa54ff53a, \\ &H_4 = 0x510e527f, H_5 = 0x9b05688c, \\ &H_6 = 0x1f83d9ab, H_7 = 0x5be0cd19 \end{aligned} \quad (32)$$

4. **Message Schedule:** For each 512-bit block, a message schedule consisting of 64 32-bit words is created. The first 16 words are the original block words, and the remaining 48 words are generated using the following formula:

$$W_t = \begin{cases} M_t \\ \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16} \end{cases} \quad (33)$$

where, $\sigma_0(x) = (x \gg 7) \oplus (x \gg 18) \oplus (x \gg 3)$ and $\sigma_1(x) = (x \gg 17) \oplus (x \gg 19) \oplus (x \gg 10)$

5. **Compression Function:** The core of SHA-256 is the compression function, which processes each 512-bit block with the initialized hash values. The compression function involves the following steps:

(a) Initialize Working Variables: Initialize eight working variables a, b, c, d, e, f, g, h with the current hash values $H_0, H_1, H_2, H_3, H_4, H_5, H_6, H_7$.

(b) Main Loop: Iterate through each of the 64 rounds using:

$$T_1 = h + \sum_1(e) + Ch(e, f, g) + K_t + W_t \quad (34)$$

$$T_2 = \sum_0(a) + Maj(a, b, c) \quad (35)$$

$$h = g, g = f, f = e, e = d + T_i \quad (36)$$

$$d = c, c = b, b = a, a = T_1 + T_2 \quad (37)$$

where,

$$\sum_0(x) = (x \gg 2) \oplus (x \gg 13) \oplus (x \gg 22) \quad (38)$$

$$\sum_1(x) = (x \gg 6) \oplus (x \gg 11) \oplus (x \gg 25) \quad (39)$$

$$Ch(x, y, z) = (x \wedge y) \oplus (\neg x \wedge z) \quad (40)$$

$$Maj(x, y, z) = (x \wedge y) \oplus (x \wedge z) \oplus (y \wedge z) \quad (41)$$

(c) Update Hash Values: After completing the rounds for each block, update the hash values:

$$H_0 = H_0 + a, H_1 = H_1 + b, H_2 = H_2 + c, H_3 = H_3 + d$$
$$H_4 = H_4 + e, H_5 = H_5 + f, H_6 = H_6 + g, H_7 = H_7 + h \quad (42)$$

6. **Final Hash Value:** After processing all blocks, the final hash value is the concatenation of the updated hash values:

$$\text{Hash} = H_0 \parallel H_1 \parallel H_2 \parallel H_3 \parallel H_4 \parallel H_5 \parallel H_6 \parallel H_7 \quad (43)$$

This 256-bit hash value serves as the output of the SHA-256 algorithm.

A.7. Steganography

Steganography [8] is the practice of concealing information within another non-secret medium to prevent detection. The word derives from the Greek words "steganos", meaning covered, and "graphein", meaning writing.

A.8. LSB-based steganography

Most commonly used technique to hide image data within image data. LSB-based steganography [52] modifies the last bit of a pixel's RGB value to embed a single bit of the secret data.

$$Pixel(i, j) = (R_{i,j}, G_{i,j}, B_{i,j}) \quad \text{Original Pixel Value.} \quad (44)$$

$$Pixel'(i, j) = (R'_{i,j}, G'_{i,j}, B'_{i,j}) \quad \text{Modified Pixel Value.} \quad (45)$$

A.9. Deep steganography

Deep Learning based Steganography [53] has the potential to create lossless images. LSB-based steganography will always have some pixel altered, but deep steganography uses Convolution Neural Networks to hide images within images. This method distributes the secret image's representation across all available bits of the cover image. This disperses the hidden information, making it less detectable.

$$L(c, c', s, s') = \parallel c - c' \parallel + \beta \parallel s - s' \parallel \quad (46)$$

Here, c and s are the cover and secret images, while c' and s' are the reconstructed versions. L is the loss function, and β controls the trade-off.

Data availability

Data will be made available on request.

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